

Voice AI Agent — Operating Cost Analysis

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FX reference: USD → INR @ ₹95.26 (market rate, 11 Jun 2026) | Pricing basis: Public list / pay-as-you-go rates, June 2026

1. Executive Summary

This report sets out the full operating cost of the AI voice-calling agent — covering the AI / voice processing stack, AWS cloud infrastructure, and telephony.

- AI + voice cost per minute:** ₹4.88 on the current model, reducing to ₹3.11 on the optimised model (gpt-realtime-mini) — a ~36% unit-cost reduction.
- AWS infrastructure:** a fixed ~₹1,400 / month, which becomes negligible per-minute as call volume grows.
- Telephony (Vobiz):** a per-minute pass-through, typically the largest single variable; shown parametrically pending the carrier's confirmed rate, with an illustrative all-in worked example in §7–8.
- Indicative all-in cost** (incl. illustrative ₹0.50/min telephony, moderate volume): ₹5.66/min current → ₹3.89/min optimised.

2. Scope & Architecture

The platform places outbound AI voice calls and conducts a scripted, interactive conversation. Each minute of call time consumes the following metered services:

Layer	Provider / Service	Role
Dialogue & speech understanding	OpenAI Realtime API (gpt-realtime)	Understands caller audio, generates the agent's replies as text
Speech transcription	OpenAI Whisper-1	Produces the caller-side transcript for call records
Speech synthesis (voice)	ElevenLabs Flash v2.5	Converts the agent's text into natural speech
Cloud hosting	AWS EC2 (ap-south-1, Mumbai)	Runs the application backend + web dashboard
Telephony	Vobiz	Carries the call over the phone network

3. Unit Cost — Per Minute of Call

Component	Rate (list)	Current gpt-realtime	Optimised gpt-realtime-mini
OpenAI — audio input	\$32 / \$10 per 1M tok	₹1.83	₹0.57
OpenAI — text output	\$24 / \$2.40 per 1M tok	₹0.57	₹0.06
Whisper-1 transcription	\$0.006 / min	₹0.57	₹0.57
ElevenLabs Flash v2.5	\$0.05 / 1k chars	₹1.90	₹1.90
AI + voice sub-total / min		₹4.88	₹3.11

Assumptions: ~600 audio-input tokens/min, ~250 text-output tokens/min, ~400 characters of synthesised speech/min.
Usage-based; varies with conversation length and verbosity. USD equivalents: \$0.051/min (current), \$0.033/min (optimised).

4. AWS Cloud Infrastructure — Fixed Monthly

Item	Basis	USD / mo	INR / mo
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EC2 compute (t2.micro)	~\$0.0124/hr × 730 hr	\$9.05	₹862
EBS storage (20 GB, gp3)	~\$0.0924 / GB-mo	\$1.85	₹176
Public IPv4 address	\$0.005/hr (AWS, since 2024)	\$3.65	₹348
Data transfer out	~64 kbps audio; minimal	~\$0.10–1.00	₹10–95
AWS total		~\$14.7	~₹1,400

Instance: t2.micro (1 vCPU, 1 GB RAM, 20 GB EBS), ap-south-1 (Mumbai), on-demand Linux. AWS is a fixed cost, independent of call volume; per-minute it falls sharply as usage rises (see §6). Free-tier eligibility, if any, is excluded as it is time-limited.

5. Telephony (Vobiz) — Pass-Through Variable

Vobiz is billed per connected minute at a rate set by the carrier and destination. As the confirmed rate is pending, the table shows the all-in impact at representative rates:

Vobiz rate	Added cost / min	Impact on a 2.5-min call
₹0.25 / min	₹0.25	₹0.63
₹0.50 / min	₹0.50	₹1.25
₹1.00 / min	₹1.00	₹2.50
₹2.00 / min	₹2.00	₹5.00

6. Blended Cost per Minute, by Volume (excl. telephony)

Monthly volume	AWS / min	Current / min	Optimised / min
1,250 min (~500 calls)	₹1.12	₹6.00	₹4.23
5,000 min (~2,000 calls)	₹0.28	₹5.16	₹3.39
20,000 min (~8,000 calls)	₹0.07	₹4.95	₹3.18

Call estimates assume an average call length of 2.5 minutes.

7. Indicative All-In Cost (incl. telephony)

Worked example at an illustrative Vobiz rate of ₹0.50/min — replace with the confirmed carrier rate for the final figure:

Monthly volume	AI + AWS / min	+ Vobiz (₹0.50)	All-in / min	All-in / 2.5-min call
1,250 min	₹6.00	₹0.50	₹6.50	₹16.25
5,000 min	₹5.16	₹0.50	₹5.66	₹14.15
20,000 min	₹4.95	₹0.50	₹5.45	₹13.63

Optimised model (gpt-realtime-mini) all-in /min at ₹0.50 Vobiz: ₹4.73 (1,250 min), ₹3.89 (5,000 min), ₹3.68 (20,000 min).

8. Monthly Cost Projection

Scenario	Minutes/mo	Current (AI+AWS)	Optimised (AI+AWS)	All-in current* (incl. Vobiz ₹0.50)
Pilot	1,250	₹7,500	₹5,288	₹8,125
Growth	5,000	₹25,800	₹16,950	₹28,300

Scale	20,000	₹99,000	₹63,600	₹1,09,000
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**Illustrative all-in uses Vobiz @ ₹0.50/min on the current model; replace with the confirmed carrier rate. Add telephony = (Vobiz rate × total minutes).*

9. Cost-Optimisation Roadmap

Lever	Est. saving	Trade-off
Switch to gpt-realtime-mini	₹1.77 / min (~36%)	Slightly lower reasoning; validate on scripted flow
Reduce average call duration	Proportional, all costs	Tighter scripting (also cuts telephony)
Disable caller transcription (if unused)	₹0.57 / min	Loses caller-side transcript records
ElevenLabs subscription tier (at volume)	Lower per-char rate	Monthly commitment
Optimise prompt caching	Marginal, free	None

10. Assumptions & Disclaimers

1. Token / character consumption is usage-based and varies with conversation length and style; figures use moderate-conversation averages.
2. AI provider rates are public list / pay-as-you-go (June 2026); volume or committed-use discounts may apply.
3. AWS instance type inferred from server specs (1 vCPU / 1 GB → t2.micro); free-tier eligibility excluded as time-limited.
4. Telephony (Vobiz) confirmed rate pending; all-in figures use an illustrative ₹0.50/min and must be updated with the carrier rate.
5. FX rate as stated; INR figures move with the exchange rate.